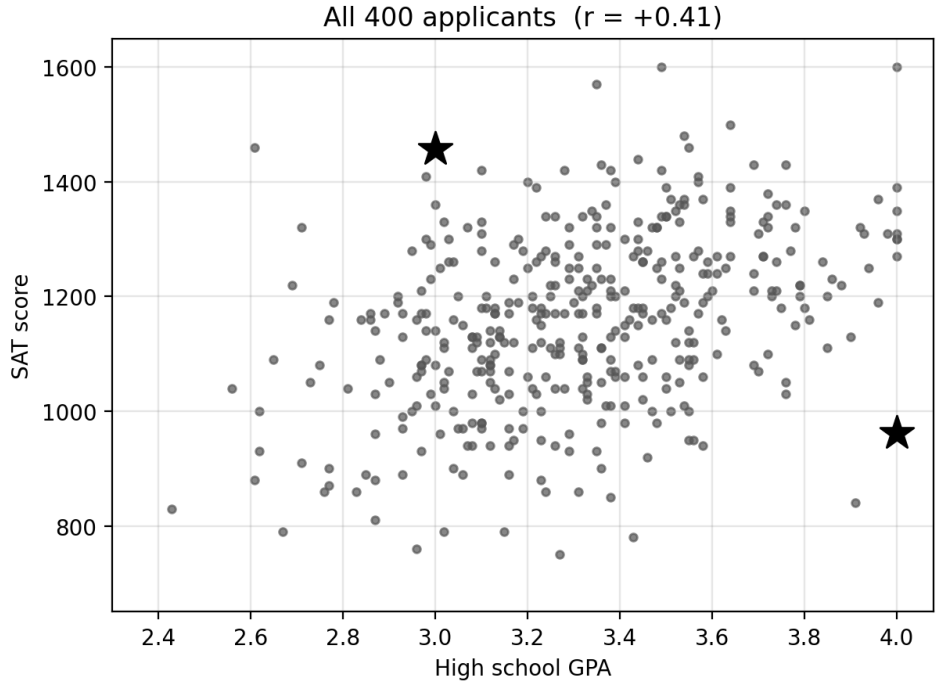
Class 20: Who Got In

What a selected group can tell you -- and what it can't

Team Members: _____ Date: _____

An honors program received 400 applications. Each dot below is one applicant's high school GPA and SAT score. Across **all** applicants, r = +0.41: higher GPAs tend to go with higher SAT scores, though far from perfectly. (The data are synthetic, built to resemble a realistic applicant pool.)

The program ranks applicants on a combined score – GPA in standard units **plus** SAT in standard units – and admits the top 30% (120 students). Every applicant exactly on the admission cutoff has the same combined score. The two stars sit on that cutoff.



> **Q1. Connect the two stars with a straight line and shade the region of admitted students. In one sentence: why does the cutoff slope downward?**

Q2 – Predict first. Among the 120 admitted students only, what will r be? Circle one, then explain using your shaded region.

strongly negative weakly negative near 0 weakly positive about +0.41 stronger than +0.41

Q3. A dean studies only the enrolled honors students and reports: "Our strongest test-takers have the weakest grades. Test-taking ability and hard work must be different traits." Using your sketch, explain what the dean got wrong.

Q4 – Predict first. Program B admits the top 30% by GPA alone, ignoring SAT. Among Program B's admits, will r be negative, near 0, or positive? Larger or smaller than +0.41?

Q5. You are all in an honors program. Name one conclusion someone could wrongly draw by studying only students like you.